

Lecture 7: Expansion at European scale

Energy Economics of the Green Transition

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Today

1. The model of ESM's section 8
2. Part 1: the baseline
3. Part 2: electric vehicles
4. Part 3: hydrogen

While we start: open the Model Lab with your code from lecture 6, *championape-pi.tailf1dc5a.ts.net*.



The model of ESM’s section 8

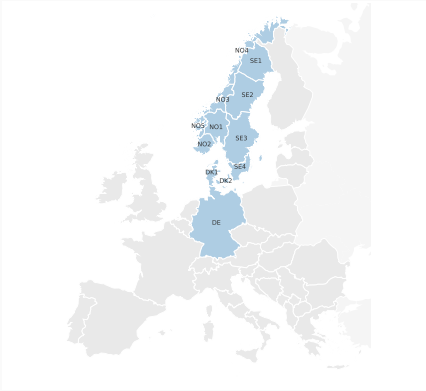
From section 7 to section 8

- **Section 7:** Denmark built from scratch, at a carbon price of 85 €/t.
- **Section 8: today's system.** Every zone keeps the plants it has today and builds beside them; interconnectors can be bought.
- **Emissions are cut** in all twelve zones together, to a budget.
- **Two new sources of demand:** electric vehicles and a hydrogen sector.

Today, we read what it costs to cut the emissions of today's power system, and what cars and hydrogen do to that cost.

The model: four scenarios

- **The question:** what should the twelve zones build beside today's plants to cut emissions, at the lowest cost?
- **Baseline** (part 1): plants, storage and interconnectors to choose.
- **Smart charging, vehicle-to-grid** (part 2): cars that choose when to charge, and may feed power back.
- **Hydrogen** (part 3): a demand for hydrogen, and the means to make, store and burn it.



Energy System Models, Fig. 6.1: the twelve zones of the model.

The Lab and the Explorer

- **The Model Lab:** lecture 7, in three parts. Your code from lecture 6 still works.
- **The Model Explorer:** the model *Decarbonising today's power system*, every scenario solved in advance. New tabs: Prices and margins, and Hydrogen.
- **Each question names its view:** tab, figure, scenario, and the scenario to compare it with.



championape-pi.tailf1dc5a.ts.net

championape.github.io/EnergyEconomics-explorer. On paper: the worksheet.

Part 1

Setup: budgets, cuts and the carbon price

- **A budget** caps the emissions of all twelve zones, as a share of what they would emit without one: a budget of 1 % is a cut of 99 %.
- **The carbon price** τ is the budget's shadow price: what the last tonne allowed costs, paid by every emitter.
- **One instrument, not two:** a tax at τ would deliver the same cut.

The model: one new choice, interconnectors

$$\max \quad W^{\text{exp}} \equiv W^{\text{inv}} - \sum_k F_k \bar{f}_k$$

$0 \le f_{k,h} \le \bar{f}_k$	$(\underline{\omega}_{k,h}, \overline{\omega}_{k,h})$	$\forall k, h$
$0 \le \bar{f}_k \le \bar{f}_k^{\max}$	$(\underline{\beta}_k, \overline{\beta}_k)$	$\forall k$

- **A candidate interconnector** k carries $f_{k,h}$ from zone a_k to zone b_k, where $\eta_k f_{k,h}$ arrives.
- **Its capacity** $\bar{f}_k$ costs F_k a MW a year, up to the size of the project.

The model: where an interconnector enters the Lagrangian

$$\mathcal{L}_k^{\text{int}} = \sum_h \Big[\underbrace{f_{k,h}\big(\eta_k \lambda_{b_k,h} - \lambda_{a_k,h}\big)}_{\text{balance, both ends}} + \underbrace{\underline{\omega}_{k,h} f_{k,h} + \overline{\omega}_{k,h}\big(\bar{f}_k - f_{k,h}\big)}_{\text{flow limits}} \Big]$$

$$\underbrace{- F_k \bar{f}_k}_{\text{fixed cost}} + \underbrace{\underline{\beta}_k \bar{f}_k + \overline{\beta}_k\big(\bar{f}_k^{\max} - \bar{f}_k\big)}_{\text{ceiling}}$$

The baseline: prices and the last quarter

1. Open the lecture notes: *Energy System Models*, section 8.2.
2. Open the Model Explorer, the model *Decarbonising today's power system*: `championape.github.io/EnergyEconomics-explorer`
3. Open the Model Lab: `championape-pi.tailf1dc5a.ts.net`

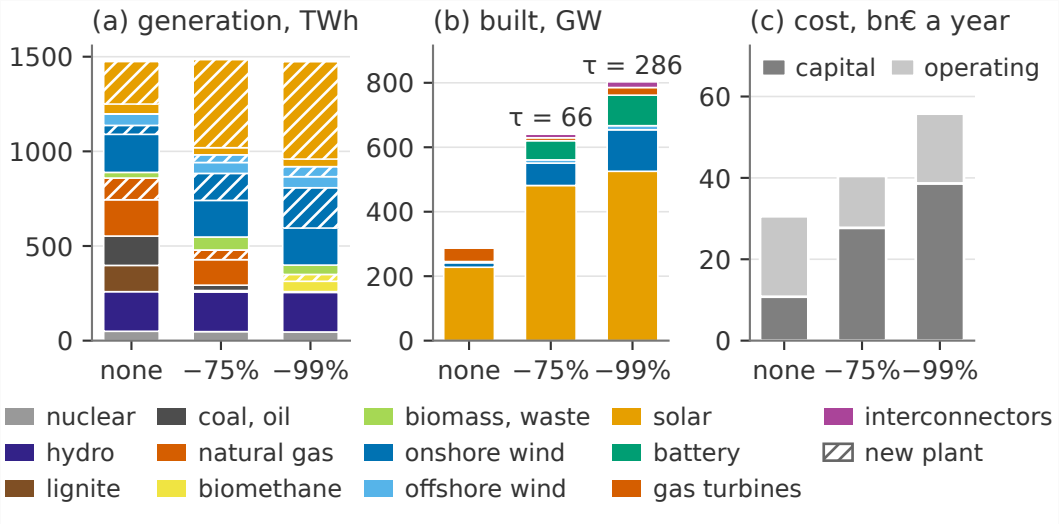
Solutions 1

Solutions 1, Q1–Q2: the questions

Q1. What the cut changes: from no cut to 75 % against 75 to 99 % (Figs. 8.1–8.2); why the price distribution changes.

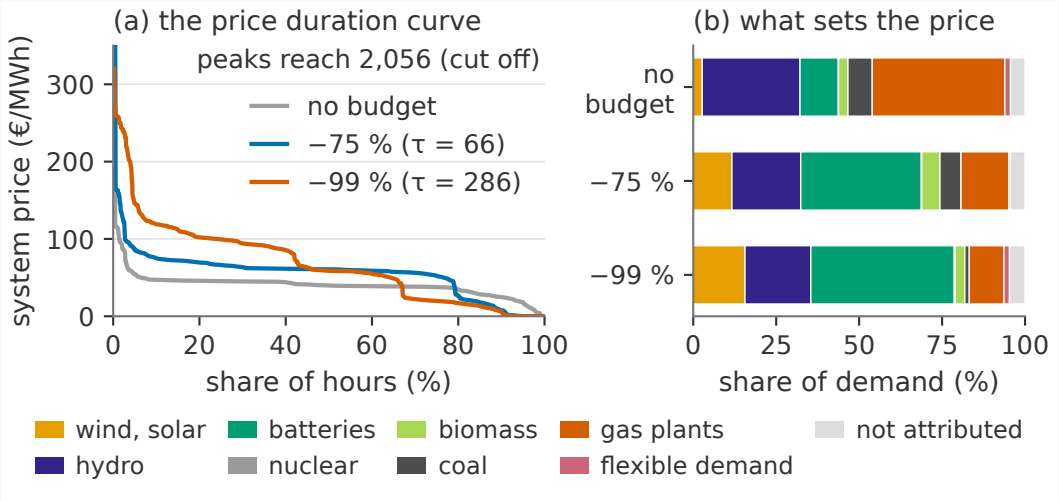
Q2. What covers the last quarter: the value of storage that holds days or weeks; why the flexible demand block steps back so much more often at a cut of 90 % than at 75 %.

Solutions 1, Q1: the first three quarters, and the last



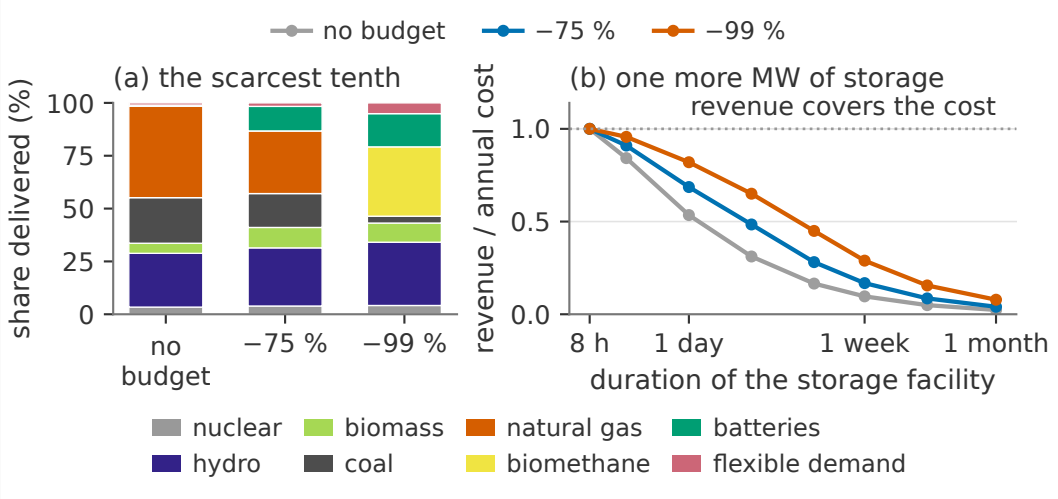
Source: ESM Fig. 8.1, the baseline without a budget and at cuts of 75 and 99 %; τ in €/t.

Solutions 1, Q1: both tails of the price distribution fatten



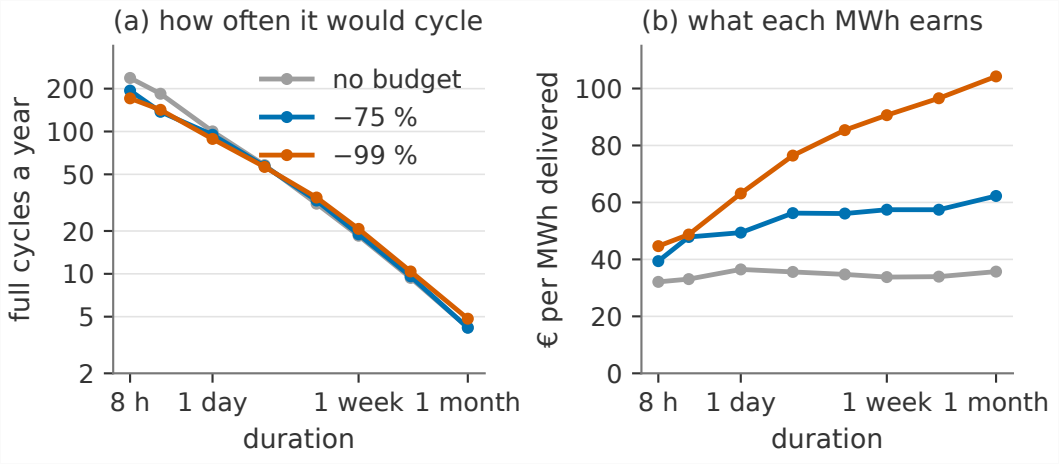
Source: ESM Fig. 8.2: (a) each hour's zonal prices weighted by the zones' demand; (b) as shares of the year's demand.

Solutions 1, Q2: long storage is worth more, and still does not pay



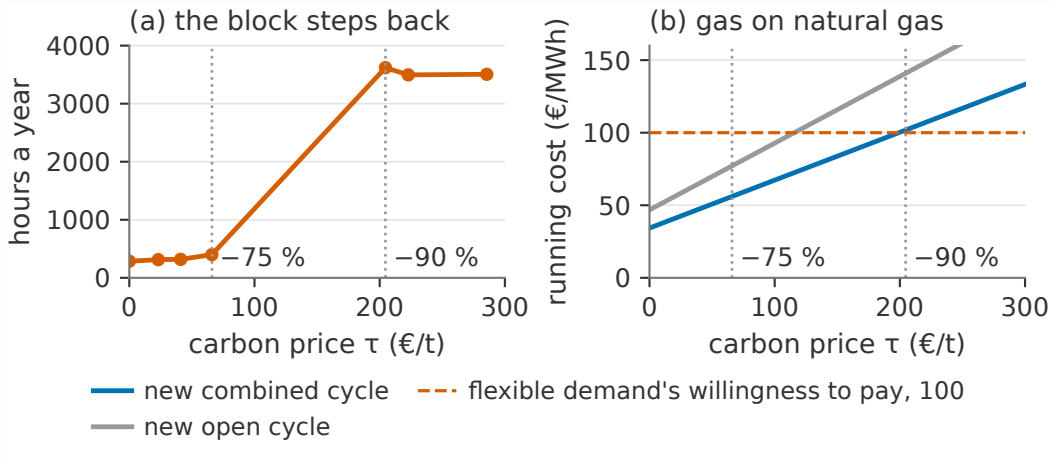
Source: ESM Fig. 8.3: (a) the tenth of the year with the highest residual demand; (b) in the zone where it earns the most.

Solutions 1, Q2: the cut changes what each MWh earns



Source: Explorer (Prices and margins > How often storage would cycle, and what each MWh would earn), at the three budgets.

Solutions 1, Q2: the carbon price passes the flexible block



Drawn from the note's results: (a) the Explorer's Sweeps, Flexible demand steps back, each budget at its carbon price; (b) the running costs the question gives.

Part 2

Electric vehicles: half a battery and a whole one

1. Section 8.3 adds the cars: with smart charging they choose when to charge; with vehicle-to-grid they may also feed power back.
2. Both at a cut of 99 %, against the baseline: compare the scenarios in the Explorer.

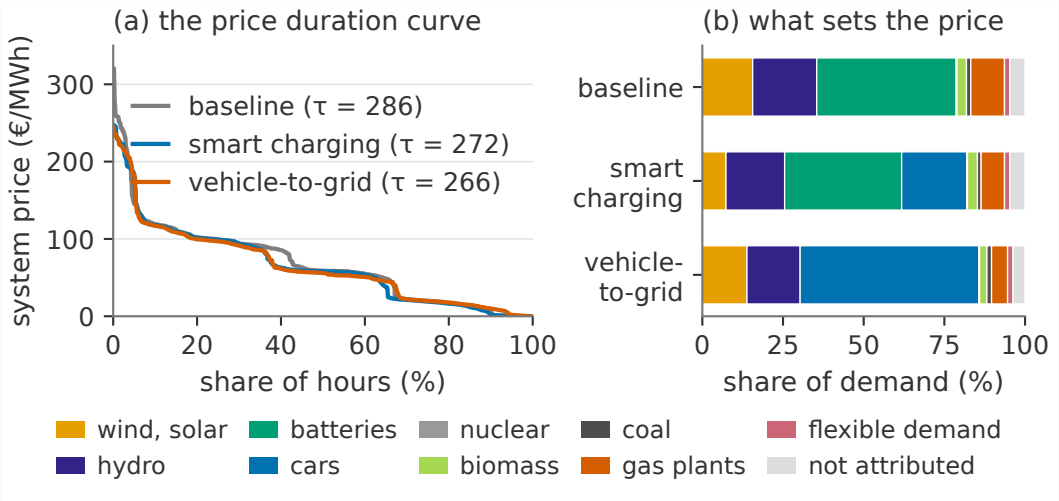
Solutions 2

Solutions 2, Q1–Q2: the questions

Q1. What the cars do to prices: compare and explain against the baseline (Fig. 8.4); what fixes the price when the cars are at the margin.

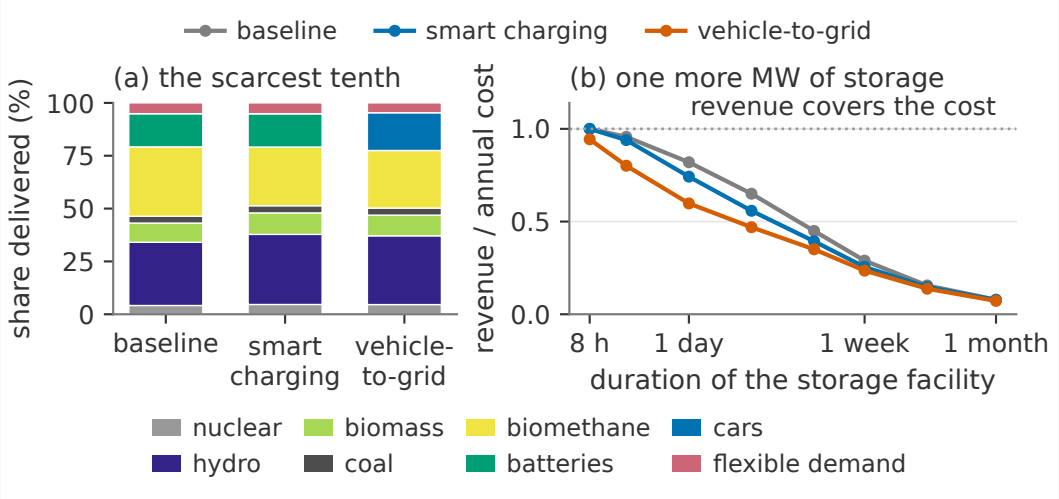
Q2. Cars against batteries: why eight-hour storage still pays with smart charging but not with vehicle-to-grid (Fig. 8.5(b)); why the cars take so little from long storage.

Solutions 2, Q1: what the cars do to the two tails



Source: ESM Fig. 8.4, the three scenarios at a cut of 99 %.

Solutions 2, Q2: the cars take the day's work, not the week's



Source: ESM Fig. 8.5, the three scenarios at a cut of 99 %: (a) the tenth of the year with the highest residual demand.

Solutions 2, Q2: the battery's investment condition

$$\overline{\beta}_s - \underline{\beta}_s = \underbrace{\sum_h \left(\overline{\phi}^c_{s,h} + \overline{\phi}^d_{s,h}\right) + \bar{\delta}_s \sum_h \overline{\psi}_{s,h}}_{\text{a MW's rents on the price spreads}} - \underbrace{F_s}_{\text{its annual fixed cost}}$$

Smart charging: batteries are built

$\underline{\beta}_s = \overline{\beta}_s = 0$

the daily spread just pays a battery; the cars replace batteries

Vehicle-to-grid: none is built

$\underline{\beta}_s > 0$

nothing holds the spread up; the cars lift the cheapest hours

Part 3

Setup: a market for hydrogen

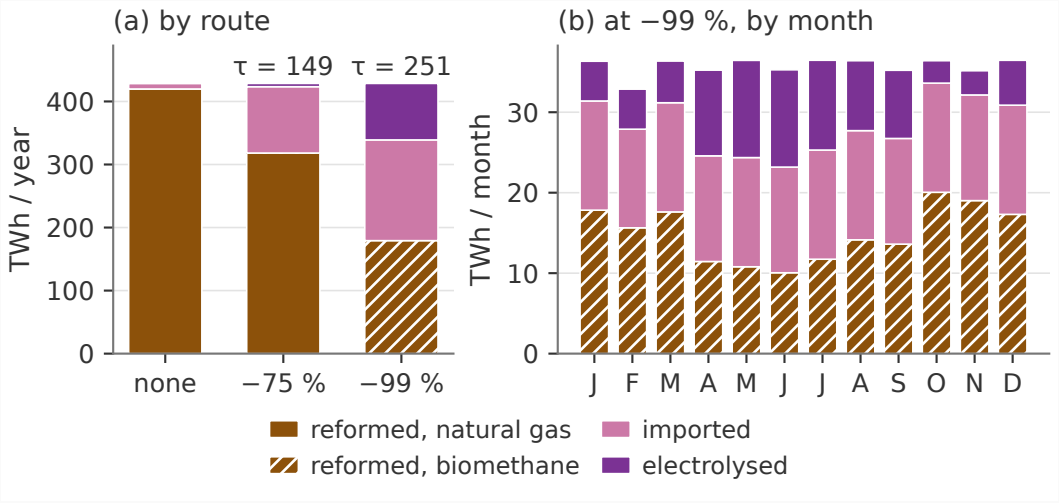
- **The hydrogen scenario** keeps smart charging and adds an industrial demand for hydrogen in Germany, Denmark and Sweden, the same in every hour.
- **Hydrogen is made** by reformers, from natural gas or biomethane; **imported** into Germany; or **electrolysed** from power.
- **It can be stored** in salt caverns or tanks, and **burnt** back into power by turbines.

Hydrogen: a flexible load, not a storage

1. The hydrogen sector, part by part, against the heat sector of section 5.
2. Electrolysers as a flexible load.

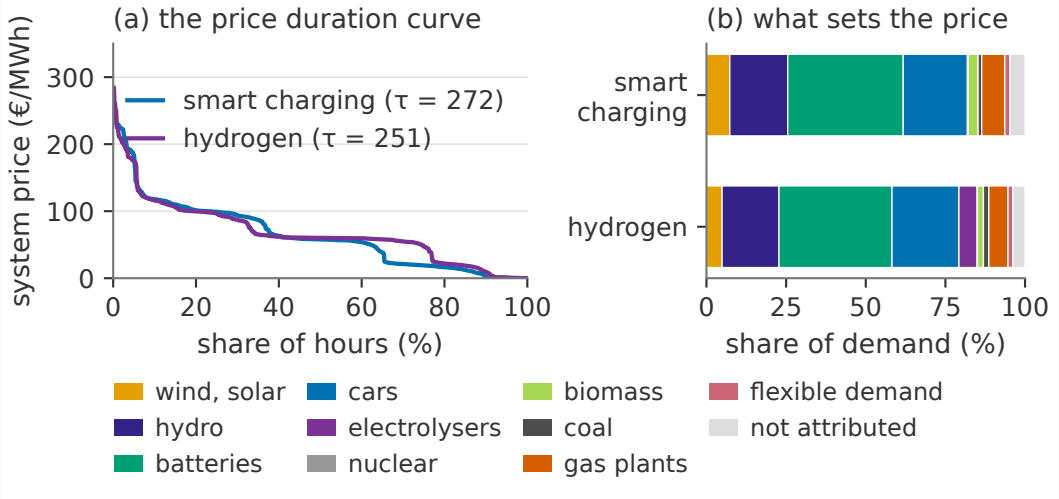
Solutions 3

Solutions 3, Q1: biomethane does the seasonal work



Source: ESM Fig. 8.7, hydrogen supply in the hydrogen scenario; (b) summed over the zones.

Solutions 3, Q2: electrolysers set the price at their hydrogen's value



Source: ESM Fig. 8.9, smart charging and the hydrogen scenario at a cut of 99 %; their budgets differ in tonnes.

Wrap-up

What carries forward

1. **The last quarter of the cut costs the most.** It is about the hours with little wind and sun, and every margin that could cover them runs into a limit of its own.
2. **Cars take over the batteries' daily work,** but they do not carry energy across weeks.
3. **Hydrogen is a flexible load, not a storage:** biomethane does the seasonal work that Victoria et al. (2019) give to hydrogen.

Model answers to today's open questions: in the Lab tomorrow. The written solutions follow.

References i

Victoria, Marta, Kun Zhu, Tom Brown, Gorm B. Andresen, and Martin Greiner (2019). **“The Role of Storage Technologies Throughout the Decarbonisation of the Sector-Coupled European Energy System”.** *Energy Conversion and Management* 201, p. 111977.